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Hierarchical fusion and local-aware transformer for occluded person re-identification

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ABSTRACT

Occluded person re-identification (ReID) is intended to address the problem of matching pedestrians when images of individuals are partially occluded. Recently, Transformer-based methods for occluded person ReID have received considerable attention. However, although existing methods have achieved promising results, most of them do not fully consider the varying contributions of different patches to identity recognition, nor do they sufficiently emphasize the identity information in critical regions. Furthermore, those methods often lack sufficient capability to extract fine-grained local features, making it difficult to fully explore the identity information embedded in various body parts. To resolve the mentioned problems, we propose a Hierarchical Fusion and Local-aware Transformer (HFLAT) for occluded person ReID. Specifically, we firstly design a feature hierarchical fusion module that hierarchizes and fuses the patch feature vector sequence according to the relative importance of each patch to the global feature vector, thereby reinforcing the identity discriminative features of key regions. We then design a feature separation module to distinguish foreground features from background features by employing patch-level saliency analysis, thereby mitigating the negative impact of backgrounds and occlusions on the performance of the model. In addition, we design a local feature extraction module which restricts the range of interactions between the features using a local-aware multi-head attention mechanism, increasing the model's ability to obtain fine-grained local features. Experimental results on the Occluded-DukeMTMC, Occluded-ReID, Market1501, and DukeMTMC-ReID datasets demonstrate that HFLAT reaches the current state-of-the-art performance for occluded person ReID. Specifically, on the Occluded-DukeMTMC and Occluded-ReID datasets, our method achieves the Rank-1 accuracy of 79.6% and 89.8%, respectively, and the mAP of 64.7% and 84.9%, respectively. On the Market1501 and DukeMTMC-ReID datasets, our method achieves the Rank-1 accuracy of 95.9% and 90.6%, respectively, and the mAP of 90.8% and 82.2%, respectively.

1. Introduction

Person re-identification (ReID) tasks intend to identify and match target individuals among multiple non-overlapping camera viewpoints. Owing to the rapid growth of deep learning techniques, research about person ReID (Luo et al., 2019; Qian et al., 2017; Zhou et al., 2019) has made significant progress. However, in practical situations, as illustrated in Fig. 1, pedestrians often exhibit missing body parts due to occlusions caused by objects such as trees and vehicles. Therefore, how to enhance the performance of person ReID models when pedestrians are partially occluded has become one of the core challenges.

In recent times, occluded person ReID has garnered heightened interest, and its key is extracting features from partially occluded images to mitigate the impact of occlusions. To this end, various occluded per-

son ReID methods have been presented. Existing methods can be broadly categorized into external auxiliary information-based methods Gao et al. (2020), Huang et al. (2020), Miao et al. (2019) and Transformer-based methods Li et al. (2021), Lin et al. (2024), Wang et al. (2022b). The external auxiliary information-based methods typically reduce the effect of occlusions on the discriminative capability of pedestrian features by locating un-occluded body regions with the help of auxiliary information, such as human posture or body key points. The Transformer-based methods primarily leverage the global modelling strengths of Transformers, integrating various attention mechanisms to extract more discriminative features. Although these methods have achieved some encouraging results, most of them process different image patches indiscriminately. Not only they fail to quantify the different contributions of different image patches to identity recognition, but they also neglect to focus on

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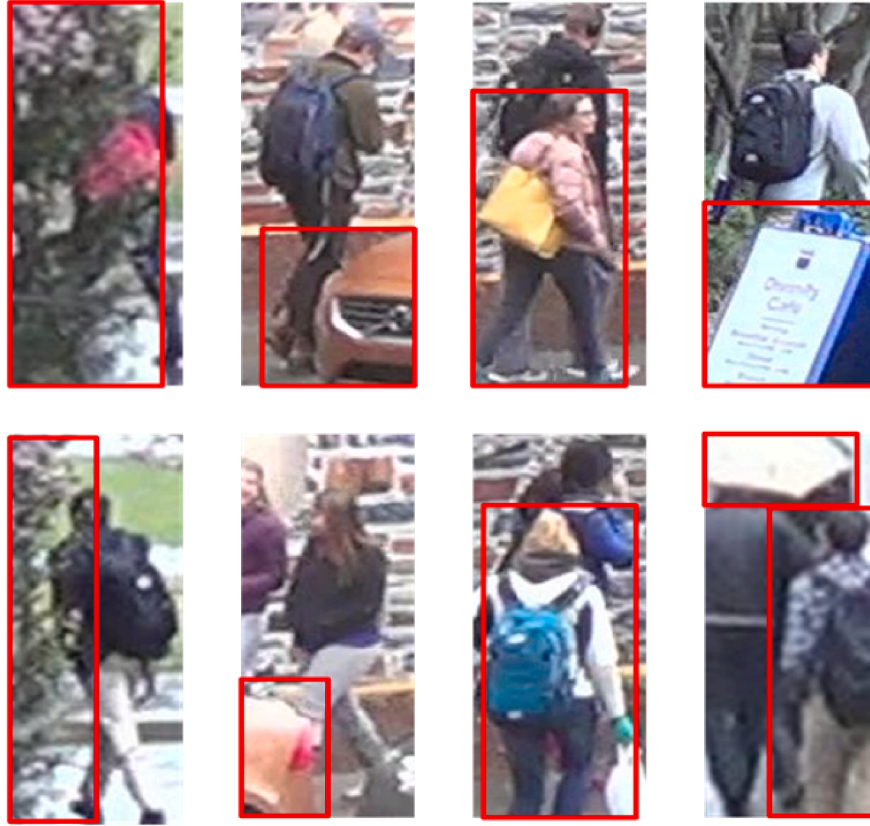


Fig. 1. Examples of pedestrians being occluded.

key regions. Moreover, most of them fail to effectively isolate invalid or detrimental information contained in pedestrian images, thereby being unable to mitigate the adverse effects of backgrounds and occlusions on the performance of their models. In particular, although most of them possess a strong ability to capture global contextual information, their capacity for acquiring fine-grained local features is largely inadequate, preventing them from fully extracting the identity information contained in different body parts of pedestrians.

To resolve the mentioned problems, we propose a hierarchical fusion and local-aware Transformer (HFLAT) for occluded person ReID. First, we design a feature hierarchical fusion module (FHFMM) to hierarchize and fuse the patch feature vector sequence based on their relative importance to the global feature vector, enhancing the identity discriminative features of key regions. Subsequently, we design a feature separation module (FSM) to isolate foreground and background features using patch-level saliency analysis, thereby mitigating the negative effects of backgrounds and occlusions on the model's performance. Additionally, we design a local feature extraction module (LFEM) that employs a local-aware multi-attention mechanism to limit feature interactions, improving the model's fine-grained local feature extraction capability. The main contributions of this research work presented in this paper are as follows:

- We propose a hierarchical fusion and local-aware Transformer (HFLAT) for occluded person ReID.
- We design a feature hierarchical fusion module (FHFMM) that hierarchizes and fuses the patch feature vectors based on their relative importance to the global feature vector, aiming to enhance the identity discriminative features of key regions.
- We design a feature separation module (FSM) that separates foreground and background features using patch-level saliency analysis to mitigate the negative effects of backgrounds and occlusions on the model's performance.

- We design a local feature extraction module (LFEM) to enhance the fine-grained local feature extraction capability of the model by limiting the range of interactions between the features using a local-aware multi-head attention mechanism.
- We perform extensive experiments on four datasets to validate the efficiency and advancement of the model.

2. Related works

As deep learning techniques progress rapidly, person ReID has achieved remarkable results [Luo et al. \(2019\)](#), [Qian et al. \(2017\)](#), [Zhou et al. \(2019\)](#). However, in practical scenarios, pedestrians in images are often partially occluded by objects such as trees or vehicles, leading to the loss of key body parts. To address this issue, scholars have presented a series of methods for occluded person ReID, which can be grouped into external auxiliary information-based methods [Gao et al. \(2020\)](#), [Huang et al. \(2020\)](#), [Miao et al. \(2019\)](#) and Transformer-based methods [Li et al. \(2021\)](#), [Lin et al. \(2024\)](#), [Wang et al. \(2022b\)](#).

The external auxiliary information-based methods primarily mitigate the impact of occlusions on their models by utilizing auxiliary information such as human posture or key points to locate visible body parts. For instance, [Miao et al. \(2019\)](#) proposed a pose-guided feature alignment (PGFA) method to align local features using human key-point information. [Gao et al. \(2020\)](#) proposed a pose-guided visible part matching (PVPMM) method that employs a pose-guided attention mechanism and a local visibility predictor to extract features from un-occluded areas accurately. [Wang et al. \(2022a\)](#) proposed a key-point-aware occlusion suppression and semantic alignment (POS) method that aligns features using human key points and employs a feature enhancement strategy to compensate for missing feature regions. [Cui et al. \(2025\)](#) proposed a pose-guided partial-attention network with batch information (PPBI), which effectively reduces the negative impact of occlusions on model performance by capturing the semantic relationships of key

points among different samples within each batch. Huang et al. (2020) proposed a human parsing based alignment with multi-task learning (HPNet) model that enhances the features in the visible regions using body part masks generated by a human parsing task. Somers et al. (2023) proposed a body part-based (BPreID) model that extracts local features using a local attention mechanism guided by human parsed labels. Dou et al. (2024) proposed a decouple re-identification and human parsing (DROP) method, which decouples the features for person ReID and human parsing tasks, mitigating the feature granularity conflict between the two tasks and improving their collaborative effectiveness. Although the above methods based on external auxiliary information have achieved some promising results, their performance may be severely degraded when human body poses or key points are inaccurately estimated because they rely too much on the accuracy of the external auxiliary information.

The Transformer-based methods mainly utilize the advantages of Transformer in global modelling, combined with the attention mechanisms to obtain more discriminative features. For instance, Li et al. (2021) proposed a part-aware Transformer (PAT) by first applying the Transformer framework to occluded person ReID tasks, which improves the representation ability of local features by employing a pixel context encoder and a local prototype decoder. Lin et al. (2024) proposed a multi-level relation-aware Transformer (MLRAT), which has stronger feature extraction ability by mining the feature relationships at patch and sample levels. Wang et al. (2022b) proposed a Transformer-based pose-guided feature disentangling (PFD) method, which effectively mitigates the negative impact of occlusions by disentangling features utilizing pose information and aligning them to un-occluded regions. Yang et al. (2023) proposed a robust feature mining Transformer (RFMT) method, which combines residual Transformer layers with a global attention mechanism, improving their model's robustness in complex contexts. Wang et al. (2024) proposed a feature completion Transformer (FCFormer), which employs an occlusion instance enhancement strategy and a feature-complementary decoder to reconstruct occluded features based on neighboring un-occluded regions. Bian et al. (2024) proposed a novel occlusion-aware feature recover (OAFR) model, which uses un-occluded local features to recover missing features. Zheng et al. (2024) proposed a cascade Transformer reasoning embedded by uncertainty network (CTU) model that progressively extracts critical pedestrian features using an uncertainty-aware self-attention mechanism.

Although the above Transformer-based methods can address the problems faced by occluded person ReID to some degree, most of them fail to adequately account for the different contributions of different image patches to identity recognition, nor do they prioritize the identity information contained in key regions. For instance, the methods such as PAT (Li et al., 2021), PFD (Wang et al., 2022b), and RFMT (Yang et al., 2023) fail to adequately account for the different contributions of different image patches to identity recognition, consequently lacking focus on key regions. Moreover, most of the existing models exhibit insufficient capability in extracting fine-grained local features, thereby failing to fully exploit the identity information embedded in different body parts of pedestrians. For instance, the methods such as MLRAT (Lin et al., 2024) and RFMT (Yang et al., 2023) do not perform fine-grained local feature extraction, consequently failing to adequately mine the local information of pedestrians. They neither adequately consider the difference in the contribution of different image patches to identity recognition, nor sufficiently emphasize the identity information of key regions. Additionally, most of their fine-grained local feature extraction capabilities are insufficient to fully explore the identity information embedded in different body parts of pedestrians.

Our work is also related to some works. For example, Eliwa et al. (2024) proposed a framework that integrates Microsoft Azure cloud services with a permissioned blockchain network. After preprocessing and anonymizing the CT images uploaded by patients via mobile terminals,

the framework stores the images in Azure Blob Storage, and realizes access control exclusive to authorized specialists through blockchain smart contracts. Abd El-Hafeez et al. (2025) proposed a novel multi-scale attention model for the classification of breast cancer histopathological images, which achieves high-precision recognition by capturing discriminative features across multiple morphological scales in histopathological images. Eliwa and Abd El-Hafeez (2025c) proposed a robust deep learning framework improved upon YOLOv11 for the multi-class classification task of cervical cancer cells, which enhances the model's classification accuracy via an Attention-Guided Multi-Scale Feature Fusion (AGMS-FF) module. Eliwa and Abd El-Hafeez (2025a) conducted a rigorous comparative evaluation of five fine-tuned deep learning architectures for rice maturity classification tasks, namely YOLOv11 enhanced with an Attention-Guided Multi-Scale Feature Fusion (AGMS-FF) module, baseline YOLOv11, ResNet18, EfficientNet-B0, and MobileNetV3. Their results verify the practical value of deep learning-based computer vision systems in sustainable rice cultivation. Hassan et al. (2025b) proposed a novel DenseNet model integrated with attention mechanisms and optimized by the Nadam algorithm, which enhances the focus on pertinent features and thereby improves the model's classification accuracy under complex conditions. Eliwa and Abd El-Hafeez (2025b) proposed an improved YOLOv11 architecture for the automated classification of peripheral blood cells, which integrates a Dynamic Cross-Scale Context Aggregation (DCSCA) module. Through parallel convolution, dynamic attention, and cross-scale interaction, the module enables multi-scale feature capture, scale recalibration, and context modeling, thereby significantly improving the model's classification accuracy for blood cells. Hassan et al. (2025d) proposed a novel network architecture named Sign Neuro-Enhanced Dense Attention Network (SNDA), which adopts the Nadam optimizer to achieve fast convergence of the training process. Hassan et al. (2025a) presented a comprehensive review of research advances in the super-resolution field from 2015 to 2025. They pointed out that the application of attention modules and domain-specific architectures has effectively improved super-resolution performance, while also indicating that current research still suffers from limitations such as insufficient real-world scenario validation and the lack of deployment considerations. Hassan et al. (2025c) proposed a network architecture based on DenseNet121 and the Squeeze-and-Excitation (SE) attention module for fruit image classification, which improves the model's generalization ability through the application of attention mechanisms, Nadam optimization, and other data augmentation methods. Eliwa and Abd El-Hafeez (2025) proposed a plant disease detection and classification method based on a modified YOLOv11 architecture, which improves the model's performance in plant disease detection and classification through strategies such as a custom classification head and dynamic validation set splitting. El-Sayed and El-Hafeez (2012) proposed an improved edge detection method for grayscale images. Based on the Baljit and Amar algorithm, this method replaces image derivative calculation with Shannon entropy, thereby reducing CPU computation time while ensuring the robustness of edge detection quality. Saabia et al. (2019) proposed a face recognition system, which achieves efficient face recognition through five sequential steps. These steps include face detection combined with median filter preprocessing, feature extraction using Gabor filters, dimensionality reduction via Principal Component Analysis (PCA), feature selection based on the Grey Wolf Optimization (GWO) algorithm, and classification using a k-NN classifier.

3. Proposed method

This section begins with an overview of the HFLAT framework, then details the feature hierarchical fusion module (FHFM), the feature separation module (FSM), and the local feature extraction module (LFEM). Finally, the loss functions employed for model training are described.

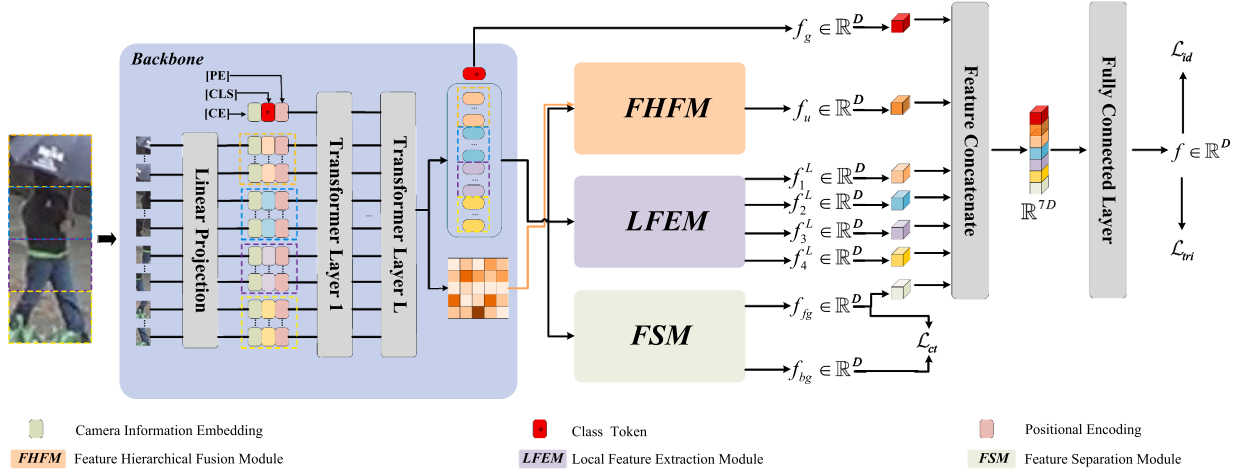


Fig. 2. Work flowchart of HFLAT. FHFMM denotes the feature hierarchical fusion module, FSM denotes the feature separation module, and LFEM denotes the local feature extraction module.

3.1. Overall

Our model utilizes the ViT as its backbone network. Specifically, given an image $X \in \mathbb{R}^{C \times H \times W}$, where C , H , and W denote the number of channels, height, and width of the image, respectively. The work flowchart of HFLAT is shown in Fig. 2. First, we use a $P \times P$ sliding window with stride S to slice X via overlapping sampling, obtaining N fixed-size patches $\{x_i \in \mathbb{R}^{C \times P \times P} | i = 1, 2, \dots, N\}$, where $N = N_H \times N_W$, $N_H = \lfloor (H + S - P)/S \rfloor$, $N_W = \lfloor (W + S - P)/S \rfloor$ and $\lfloor \cdot \rfloor$ denotes the floor operation. We then flatten each patch into a vector and use a linear projection to map that vector into the D -dimensional feature space, yielding a sequence $Z \in \mathbb{R}^{N \times D}$ containing N embeddings. Furthermore, we introduce a learnable class token $z_{cls} \in \mathbb{R}^{1 \times D}$ into Z to capture the global information in the pedestrian image. Meanwhile, we add the learnable positional embeddings $E_{pos} \in \mathbb{R}^{(N+1) \times D}$ and camera embeddings $E_{cam} \in \mathbb{R}^{(N+1) \times D}$ to Z in order to preserve the image patches' spatial position information and camera information. Finally, we obtain the input sequence $Z_0 = [z_{cls}; z_1; z_2; \dots; z_N] + E_{pos} + E_{cam}$.

We feed Z_0 into L Transformer layers to get an output feature sequence $Z_L = [z_{cls}; z_1; z_2; \dots; z_N] \in \mathbb{R}^{(N+1) \times D}$ and an attention weight matrix $M_L \in \mathbb{R}^{(1+N) \times (1+N)}$. For Z_L , we first split it into a global feature vector $z_{cls} \in \mathbb{R}^D$ and a patch feature vector sequence $Z_{patch} = [z_1; z_2; \dots; z_N] \in \mathbb{R}^{N \times D}$. Then, we feed Z_{patch} and M_L into the feature hierarchical fusion module. In this module, we hierarchize and fuse the patch feature vectors in Z_{patch} based on the relative importance of each patch to the global feature vector, thus obtaining the fused feature vector f_u . Meanwhile, we feed Z_{patch} into the feature separation module to separate the foreground features and background features and to obtain a more discriminative foreground feature vector f_{fg} by calculating the probability that each patch belongs to the foreground employing patch-level saliency analysis. Additionally, we feed Z_{patch} into the local feature extraction module, in which we insert four learnable local tokens into Z_{patch} and use the local-aware multi-head attention to limit the range of feature interactions, extracting four local feature vectors $\{f_i^L\}_{i=1}^4$. Lastly, inspired by Sun et al. (2018), Miao et al. (2019), and Xu et al. (2022), we concatenate the global feature vector z_{cls} , the fused feature vector f_u , the foreground feature vector f_{fg} , and the local feature vectors $\{f_i^L\}_{i=1}^4$. The concatenated feature vector is then processed through a fully connected layer for information interaction and dimensionality reduction, yielding the final pedestrian representation f .

3.2. Feature hierarchical fusion module

Most of current Transformer-based occluded person ReID methods fail to adequately account for the different contributions of different image patches to identity recognition, nor do they prioritize the identification information contained in key regions. To resolve these issues, we design a feature hierarchical fusion module, as illustrated in Fig. 3, which divides the patch feature vector sequence into four different levels of sub-sequences according to the relative importance of each patch to the global feature vector, and fuses these sub-sequences to enhance the identity discriminative features of key regions. That is to say, FHFMM directs greater attention towards key regions, thereby enhancing the identity information they contain by dynamically stratifying image patch feature vectors based on their relative importance to the global feature vector.

Specifically, given a patch feature vector sequence $Z_{patch} = [z_1; z_2; \dots; z_N]$ and an attention weight matrix $M_L \in \mathbb{R}^{(1+N) \times (1+N)}$, we first define a relative importance vector $I \in \mathbb{R}^N$ to measure the degree of significance of each patch in relation to the global feature vector z_{cls} :

$$I_i = \frac{M_{cls,i} \times \|z_i\|_2}{\sum_{j=1}^N M_{cls,j} \times \|z_j\|_2} \quad (i = 1, 2, \dots, N), \quad (1)$$

where $M_{cls,i}$ denotes the attention weight of the i -th patch relative to the class token in M_L . Then, we sort the N patch feature vectors $z_1, z_2, \dots, z_N$ according to I in descending order of relative importance, and divide the sorted sequence into the first-level feature vector sub-sequence $F_{top} \in \mathbb{R}^{\frac{N}{4} \times D}$, the second-level sub-sequence $F_{med} \in \mathbb{R}^{\frac{N}{4} \times D}$, the third-level sub-sequence $F_{mid} \in \mathbb{R}^{\frac{N}{4} \times D}$, and the fourth-level sub-sequence $F_{low} \in \mathbb{R}^{\frac{N}{4} \times D}$ in equal portions.

Furthermore, we perform hierarchical fusion of these four levels of feature vector sub-sequences in the following manner to obtain the fused feature sequence $F_{fus} \in \mathbb{R}^{N \times D}$:

$$\begin{cases} F_1 = F_{top} \\ F_2 = F_1 + F_{med} \\ F_3 = F_2 + F_{mid} \\ F_4 = F_3 + F_{low} \\ F_{fus} = [F_1; F_2; F_3; F_4] \end{cases} \quad (2)$$

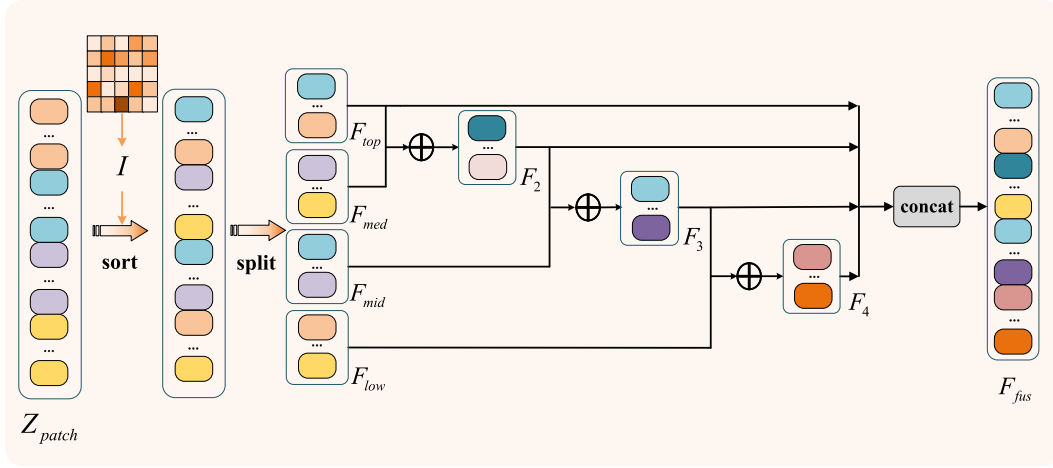


Fig. 3. Work flowchart of feature hierarchical fusion module.

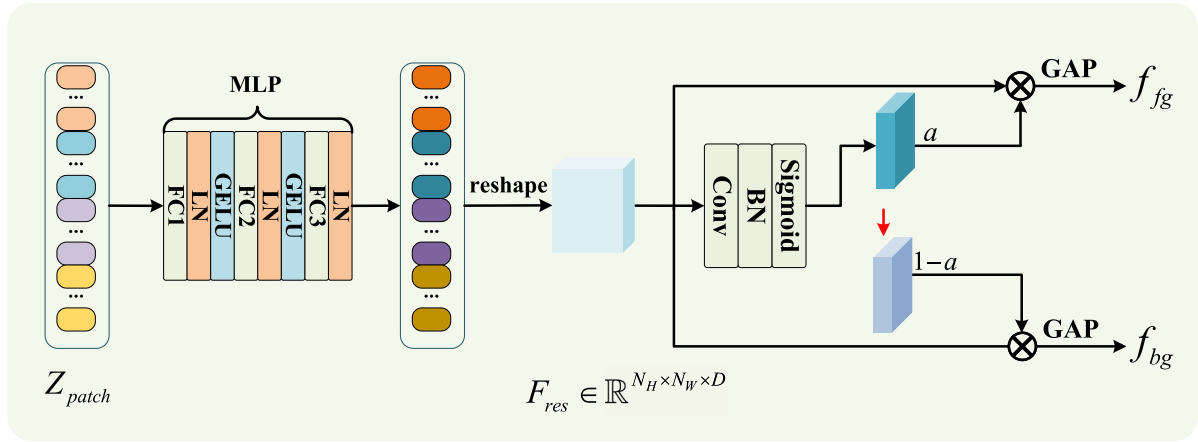


Fig. 4. Work flowchart of feature separation module.

Finally, we apply average pooling to F_{fus} along the token dimension to obtain the fusion feature vector $f_u \in \mathbb{R}^D$.

3.3. Feature separation module

In the occluded person ReID tasks, foreground features are crucial for recognizing individuals. To effectively extract the foreground features of images, as illustrated in Fig. 4, we design a feature separation module. This module employs a patch-level saliency analysis to calculate the probability that each patch belongs to the foreground, thereby achieving effective separation of foreground and background features. The core objective of this module is to enhance the discriminative ability of pedestrian representations by extracting and enhancing foreground features while suppressing interfering features caused by backgrounds and occlusions. Compared to the methods such as PGFA (Miao et al., 2019), PVP (Gao et al., 2020), and POS (Wang et al., 2022a) that rely on external auxiliary information for extracting features from un-occluded regions, this module operates independently of external auxiliary information. This eliminates the negative impact of inaccuracies in human pose or key-point estimation on the model's performance. Unlike the methods such as FCFormer (Wang et al., 2024) and OAFR (Bian et al., 2024), which reconstruct occluded features based on those from un-occluded regions, this module employs a patch-level saliency analysis to separate and enhance foreground features. This approach effectively mitigates potential errors introduced during feature reconstruction that could compromise the model's performance.

Specifically, given a patch feature vector sequence $Z_{patch} = [z_1; z_2; \dots; z_N] \in \mathbb{R}^{N \times D}$, we first process it using a multi-layer perceptron (MLP) that includes three fully connected layers, three layer normalization (LN) operations, and two GELU activation functions to obtain the feature vectors $\tilde{Z} \in \mathbb{R}^{N \times D}$:

$$\tilde{Z} = \text{LN}(\text{GELU}(\text{LN}(\text{GELU}(\text{LN}(Z_{patch}W_1 + b_1))W_2 + b_2))W_3 + b_3), \quad (3)$$

where $W_1 \in \mathbb{R}^{D \times \frac{D}{2}}$, $W_2 \in \mathbb{R}^{\frac{D}{2} \times \frac{D}{2}}$, and $W_3 \in \mathbb{R}^{\frac{D}{2} \times D}$ denote the parameters associated with the fully connected layers $FC1$, $FC2$, and $FC3$, respectively, b_1 , b_2 , and b_3 denote the corresponding bias vectors.

Then, we reshape $\tilde{Z}$ into a feature tensor $F \in \mathbb{R}^{N_H \times N_W \times D}$. Here, $N_H \times N_W = N$, and the definitions of N_H and N_W are provided in Subsection 3.1. Further, we process F with a 3×3 convolutional layer, a batch normalization (BN) layer, and apply a sigmoid function to obtain the saliency map $a \in \mathbb{R}^{N_H \times N_W}$, where $a_{i,j}$ denotes the probability that the patch at position (i, j) is part of the foreground. Furthermore, we utilize a to separate F into the foreground feature tensor $F_{fg} \in \mathbb{R}^{N_H \times N_W \times D}$ and the background feature tensor $F_{bg} \in \mathbb{R}^{N_H \times N_W \times D}$. The process mentioned above can be formalized as:

$$\begin{cases} a = \sigma(\text{BN}(\text{Conv}(F_{res}))) \\ F_{fg} = F_{res} \times a \\ F_{bg} = F_{res} \times (1 - a) \end{cases}, \quad (4)$$

where $\sigma(\cdot)$ denotes the sigmoid activation function, $\text{Conv}(\cdot)$ denotes a 3×3 convolutional layer.

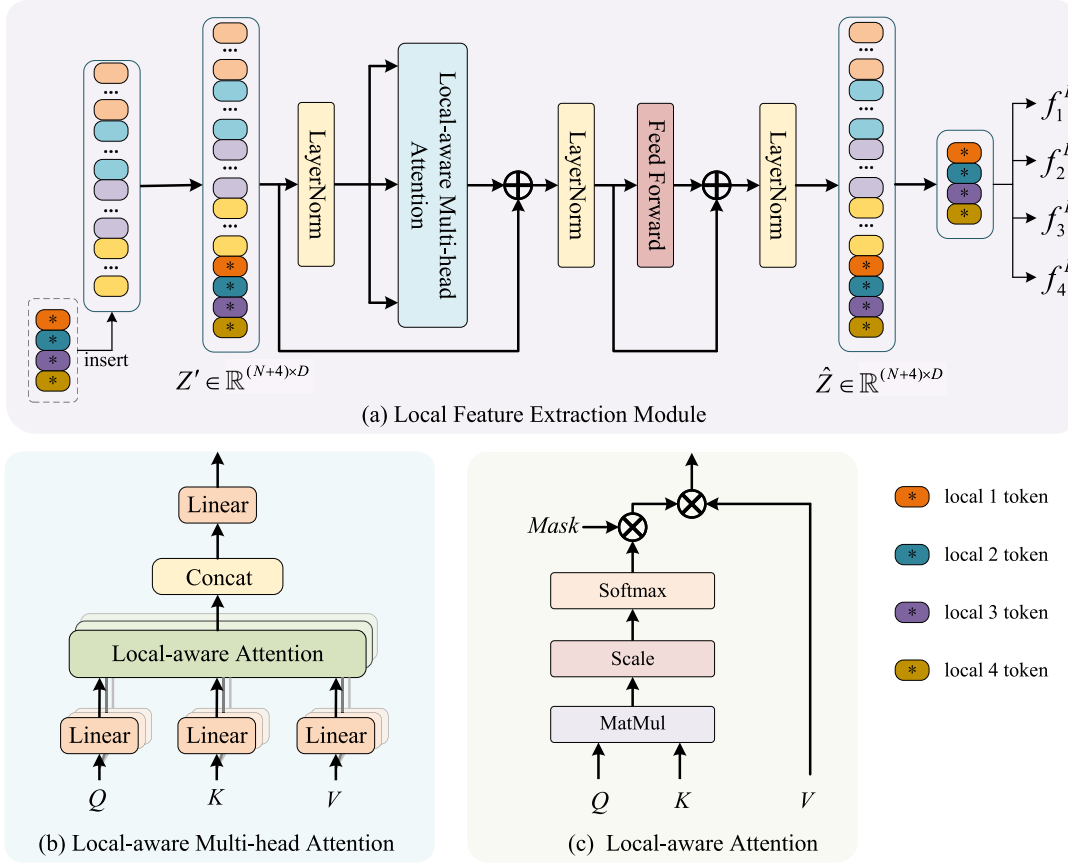


Fig. 5. Work flowcharts of (a) Local feature extraction module, (b) Local-aware multi-head attention and (c) Local-aware attention.

Finally, we apply the global average pooling to F_{fg} and F_{bg} to obtain the foreground feature vector $f_{fg} \in \mathbb{R}^D$ and the background feature vector $f_{bg} \in \mathbb{R}^D$, respectively.

3.4. Local feature extraction module

Different regions of a pedestrian image generally carry different information. For example, the upper part often contains head information, the mid-upper part often contains upper-garment information, the mid-lower part often contains pants information, and the lower part often contains shoe information. To obtain fine-grained local features in different parts of a pedestrian image, we design a local feature extraction module, as shown in Fig. 5(a). In particular, we also design and use a local-aware multi-head attention in this module, as shown in Fig. 5(b), where each attention head is a local-aware attention, as shown in Fig. 5(c). Different from the conventional multi-head self-attention, our local-aware multi-head attention achieves an effective interaction of features from a same local region by strictly restricting the interaction scope of feature vectors, which ensures the module's ability to capture fine-grained local features.

Specifically, given a patch feature sequence $Z_{patch} = [z_1; z_2; \dots; z_N] \in \mathbb{R}^{N \times D}$, we first append four learnable local tokens to its end to capture local features of the top-to-bottom four height-equally divided parts of a pedestrian image. Then, we initialize the four learnable local tokens with a zero vector. This initialization approach enables our model to learn discriminative local features directly from the data without relying on external auxiliary information (such as predefined human poses or key-points). Specifically, the feature vector sequence after adding the four local tokens can be represented as $Z' = [z_1; z_2; \dots; z_N; z_{p_1}; z_{p_2}; z_{p_3}; z_{p_4}] \in \mathbb{R}^{(N+4) \times D}$. Finally, we process Z' using a local-aware multi-head attention mechanism.

For each attention head, we first linearly transform Z' into the query Q , key K , and value V , respectively:

$$\begin{cases} Q = Z'W_q = [q_1; q_2; \dots; q_N; q_{p_1}; q_{p_2}; q_{p_3}; q_{p_4}] \\ K = Z'W_k = [k_1; k_2; \dots; k_N; k_{p_1}; k_{p_2}; k_{p_3}; k_{p_4}] \\ V = Z'W_v = [v_1; v_2; \dots; v_N; v_{p_1}; v_{p_2}; v_{p_3}; v_{p_4}] \end{cases} \quad (5)$$

where $W_q \in \mathbb{R}^{D \times D}$, $W_k \in \mathbb{R}^{D \times D}$, and $W_v \in \mathbb{R}^{D \times D}$ denote the linear transformation matrices. Then, we define a masking matrix $\text{Mask} \in \mathbb{R}^{(N+4) \times (N+4)}$ to ensure that only the patch feature vectors and a local token belonging to the same region can interact with each other for information. In particular, the elements in Mask are assigned values according to the following rules:

- When $1 \leq i \leq N$ and $1 \leq j \leq N$, if patches i and j belong to a same region, $\text{Mask}(i, j)$ is set to 1; otherwise 0.
- When $1 \leq i \leq N$ and $N+1 \leq j \leq N+4$, if patch i and local token j belong to a same region, $\text{Mask}(i, j)$ is set to 1; otherwise 0.
- When $N+1 \leq i \leq N+4$ and $1 \leq j \leq N$, if local token i and patch j belong to a same region, $\text{Mask}(i, j)$ is set to 1; otherwise 0.
- When $N+1 \leq i \leq N+4$ and $N+1 \leq j \leq N+4$, if $i = j$, $\text{Mask}(i, j)$ is set to 1; otherwise 0.

Obviously, the masking matrix Mask we define strictly restricts the interaction scope of feature vectors. That is, information exchange is permitted only between image patch feature vectors within a same local region, between image patch feature vectors and their corresponding local tokens, and between local tokens belonging to a same local region. Cross-region feature interactions are entirely prohibited. This design primarily leverages the structural characteristics of pedestrian images, wherein the discriminative ability of fine-grained local features (such as garment textures and accessory details) manifests predominantly within specific semantic regions (e.g., the textural correlation between cuffs

and the body of a garment, or the detailed correspondence between shoulders and a backpack).

Further, we calculate the output $\tilde{Z} \in \mathbb{R}^{(N+4) \times D}$ of the attention head using the following equation:

$$\tilde{Z} = \left(\text{SoftMax} \left(\frac{QK^T}{\sqrt{D}} \right) \odot \text{Mask} \right) V, \quad (6)$$

where $\odot$ denotes the matrix Hadamard product.

Furthermore, we concatenate the outputs of all the attention heads and execute a linear transformation aimed at decreasing the dimension of the concatenated result to D , obtaining the output of the local-aware multi-head attention $\hat{Z} \in \mathbb{R}^{(N+4) \times D}$. Then, as shown in Fig. 5(a), we conduct the following operations to derive the feature vector sequence $\hat{Z} \in \mathbb{R}^{(N+4) \times D}$:

$$\begin{cases} Z_m = \text{LayerNorm}(Z' + \tilde{Z}) \\ \hat{Z} = \text{LayerNorm}(Z_m + \text{FeedForward}(Z_m)) \end{cases} \quad (7)$$

where $\text{LayerNorm}(\cdot)$ denotes the layer normalization operation, and $\text{FeedForward}(\cdot)$ denotes a feedforward network consisting of two fully connected layers.

Finally, we take the four local tokens from the end of $\hat{Z} = [\hat{z}_1; \hat{z}_2; \dots; \hat{z}_N; \hat{z}_{p_1}; \hat{z}_{p_2}; \hat{z}_{p_3}; \hat{z}_{p_4}]$ and use them as the final local feature vectors $\{f_i^L\}_{i=1}^4$.

3.5. Loss functions

Total Loss: The total training loss of HFLAT includes an ID loss, a triplet loss, and a foreground-background contrastive loss, i.e.,

$$L_{\text{total}} = L_{\text{id}} + L_{\text{tri}} + \beta L_{\text{ct}}, \quad (8)$$

where L_{id} , L_{tri} , and L_{ct} respectively represent the ID loss, the triplet loss, and the foreground-background contrastive loss and β represents a balance parameter. Although we manually tune the value of β in the experiments of this study, we strongly recommend employing the hyperparameter optimization method described in Shukla et al. (2025) to determine the value of β .

ID Loss: If pedestrian samples sharing a same ID are regarded as members of a same class, then a person ReID task can be seen as an image classification task. Thus, we utilize the cross-entropy loss (Mao et al., 2023) for the ID loss, which is defined as:

$$L_{\text{id}} = -\frac{1}{N} \sum_{n=1}^N q_n \log(p_n), \quad (9)$$

where N represents the amount of training samples per mini-batch, q_n represents the n -th sample's true class probability distribution, and p_n represents the n -th sample's predicted class probability distribution.

Triplet Loss: We utilize the Batch-Hard triplet loss (Hermans et al., 2017) for the triplet loss, which is defined as:

$$L_{\text{tri}} = \sum_{i=1}^G \sum_{j=1}^Q \left[d_p^{i,j} - d_n^{i,j} + \alpha \right]_+, \quad (10)$$

where G and Q represent the amount of IDs in each mini-batch and the amount of samples with a same ID, respectively, $[g]_+$ represents the maximum value function $\max(g, 0)$, $d_p^{i,j}$ and $d_n^{i,j}$ respectively represent the Euclidean distance between positive and negative sample pairs and α represents the margin.

Foreground-background Contrastive Loss: The foreground feature vector f_{fg} and background feature vector f_{bg} output by the feature separation module should be significantly different. To ensure a high degree of similarity between the foreground feature vectors of samples with a same ID while maximizing the discrepancy between the foreground vector and the background feature vector of a same sample, we design

a foreground-background contrastive loss as follows:

$$\begin{cases} L_{fg} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(f_{fg}^i, f_{fg}^{i+})/\tau)}{\sum_{j=1}^N \exp(\text{sim}(f_{fg}^i, f_{fg}^j)/\tau)} \\ L_{fb} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(1 - \text{sim}(f_{fg}^i, f_{bg}^i)/\tau)}{\sum_{j=1}^N \exp(1 - \text{sim}(f_{fg}^j, f_{bg}^j)/\tau)} \\ L_{ct} = L_{fg} + L_{fb} \end{cases} \quad (11)$$

where N represents the quantity of training samples per mini-batch, $\text{sim}(\cdot)$ means the cosine similarity, f_{fg}^i (f_{bg}^j) means the i -th(j -th) sample's foreground feature vector, f_{fg}^{i+} means the foreground feature vector of one sample having the same ID as the i -th sample, f_{bg}^i (f_{bg}^j) means the i -th(j -th) sample's background feature vector, and τ means the distribution control factor.

4. Experiments

4.1. Datasets and evaluation metrics

To evaluate the performance of HFLAT, we assess it on the occlusion datasets Occluded-DukeMTMC (Miao et al., 2019) and Occluded-ReID (Zhuo et al., 2018), and the general datasets Market-1501 (Zheng et al., 2015) and DukeMTMC-ReID (Zheng et al., 2017). The basic information of the four datasets is presented in Table 1.

The Market-1501 dataset (Zheng et al., 2015) is released by Tsinghua University in 2015. It comprises 12,936 training images, 3368 query images, and 15,913 gallery images, captured by six cameras. Specifically, each individual is recorded by at least two different cameras.

The DukeMTMC-ReID dataset (Zheng et al., 2017) is released by Duke University in 2017. It consists of 16,522 training images, 2228 query images, and 17,661 gallery images, captured by eight cameras. In the query set and gallery set, 1404 individuals are recorded by two or more cameras, and 408 individuals are considered to be disturbing identities because they are recorded by just one camera.

The Occluded-DukeMTMC dataset (Miao et al., 2019) is based on the DukeMTMC-ReID dataset and specifically used for occluded person ReID. It comprises 15,618 training images, 2210 query images, and 17,661 gallery images. The majority of the images in the query set contain various degrees of occlusions, while the gallery set includes both complete pedestrian images and pedestrian images with occlusions, simulating a more challenging real-world surveillance environment.

The Occluded-ReID dataset (Zhuo et al., 2018) is released by Zhou et al. in 2018, designed to evaluate the performance of models under different occlusion conditions. It comprises 2000 images of 200 pedestrians, with each pedestrian including 5 full images and 5 occluded images with various degrees of occlusions. The 2000 images are divided into 1000 query images and 1000 gallery images. In accordance with the evaluation protocol adopted in existing works, Occluded-ReID is exclusively employed for testing. The models are usually trained on the training set of the Market-1501 dataset.

Following the commonly used evaluation strategy in occluded person re-identification, we employ the mean average precision (mAP) and Rank-1 accuracy to assess the performance of models.

4.2. Experimental settings

We implement our model using Python 3.8 and PyTorch 1.9, and conduct experiments on an NVIDIA 3090 GPU. The backbone is a ViT pretrained on ImageNet (Zhang et al., 2019). In our experiments, both training and testing images are resized to 256×128 pixels. Additionally, random horizontal flip, fill, random crop, and random erase strategies are employed to enhance the training images. Throughout training, we set the batch size to 64, stride S to 11, and embedding dimension D to 768. The model parameters are optimized using the Adam optimizer,

Table 1

Basic information of the four datasets.

Datasets	Number of training images (IDs)	Number of query images (IDs)	Number of gallery images (IDs)
Market-1501 (Zheng et al., 2015)	12,936(751)	3368(750)	15,913(750)
DukeMTMC-ReID (Zheng et al., 2017)	16,522(702)	2228(702)	17,661(1110)
Occluded-DukeMTMC (Miao et al., 2019)	15,618(702)	2210(519)	17,661(519)
Occluded-ReID (Zhuo et al., 2018)	–	1000(200)	1000(200)

Table 2

Values of key parameters, hardware and software requirements, and parameter count of our model.

Hyperparameter	Description	Value
Epochs	Number of training epochs	60
Batch Size	Number of samples per training batch	64
Learning Rate	Initial learning rate	8×10^{-3}
Data Augmentation	Enable or disable data augmentation	True
Inference Time (ms)	Average time per image (256×128 resolution)	2.89 ms
Parameter Count	Total number of trainable parameters	114.8 million
Hardware Requirements	GPU/CPU	NVIDIA 3090 GPU
Software Requirements	Software	Python 3.8, PyTorch 1.9
Momentum parameter	Momentum parameter of the Adam optimizer	0.9
Weight decay factor	Weight decay factor of the Adam optimizer	1×10^{-4}
Balance parameter β	Balance parameter β of the loss function	0.3
Number of levels	Number of levels in patch sequence segmentation in FHFM	4

Table 3

Experimental results(including mAP (%) and Rank-1 accuracy (%)) of HFLAT and several SOTA occluded person ReID methods on the Occluded-DukeMTMC and Occluded-ReID datasets. “–” indicates unavailable data, bold-formatted data indicates the optimal results, and underlined data indicates the sub-optimal results.

Method	Year	Occluded-DukeMTMC		Occluded-ReID	
		mAP	Rank-1	mAP	Rank-1
PGFA (Miao et al., 2019)	2019	37.3	51.4	–	–
PVPM (Gao et al., 2020)	2020	37.7	47.0	61.2	70.4
HPNet (Huang et al., 2020)	2020	–	–	77.4	87.3
POS (Wang et al., 2022a)	2022	54.0	65.0	–	–
TransReID (He et al., 2021)	2021	59.2	66.4	77.8	81.5
PAT (Li et al., 2021)	2021	53.6	64.5	72.1	81.6
PFD (Wang et al., 2022b)	2022	61.8	69.5	83.0	81.5
FED (Wang et al., 2022c)	2022	56.3	67.9	82.3	87.0
SAP (Jia et al., 2023)	2023	62.2	70.7	76.8	83.0
PRE-Net (Yan et al., 2023)	2023	55.2	68.3	–	–
CAAO (Zhao et al., 2023)	2023	59.5	68.5	83.4	87.1
RFMT (Yang et al., 2023)	2023	60.6	69.0	78.0	82.5
SSPEM (Pang et al., 2024)	2024	62.8	70.2	78.5	82.8
MLRAT (Lin et al., 2024)	2024	<u>63.1</u>	73.3	<u>83.6</u>	<u>88.9</u>
OA-ReID (Ren et al., 2025)	2025	62.4	<u>74.2</u>	81.5	87.1
DPEFormer (Zhang et al., 2025)	2025	58.9	69.9	79.5	87.0
HFLAT(ours)	2025	64.7	79.6	84.9	89.8

Table 4

Experimental results(including mAP (%) and Rank-1 accuracy (%)) of HFLAT and several SOTA occluded person ReID methods on the Market-1501 and DukeMTMC-ReID datasets. “–” indicates unavailable data, bold-formatted data indicates the optimal results, and underlined data indicates the sub-optimal results.

Method	Year	Market-1501		DukeMTMC-ReID	
		mAP	Rank-1	mAP	Rank-1
DSR (He et al., 2018a)	2018	64.3	83.6	–	–
PCB (Sun et al., 2018)	2018	77.4	92.3	66.1	81.8
PGFA (Miao et al., 2019)	2019	76.8	91.2	65.5	82.6
HOREID (Wang et al., 2020)	2020	84.9	94.2	75.6	86.9
HPNet (Wang et al., 2020)	2020	74.6	91.2	–	–
POS (Wang et al., 2022a)	2022	86.2	95.0	76.7	88.7
TransReID (He et al., 2021)	2021	88.9	95.2	82.0	90.7
PAT (Li et al., 2021)	2021	88.0	95.4	78.2	88.8
FED (Wang et al., 2022c)	2022	86.3	95.0	78.0	89.4
FRT (Xu et al., 2022)	2023	88.1	95.5	81.7	90.5
FCFormer (Wang et al., 2024)	2023	86.8	95.0	78.8	89.7
RFMT (Yang et al., 2023)	2023	89.2	95.4	82.4	90.5
CTU (Zheng et al., 2024)	2024	88.3	95.7	78.3	89.5
OA-ReID (Ren et al., 2025)	2025	89.1	<u>95.8</u>	81.2	90.4
DPEFormer (Zhang et al., 2025)	2025	88.1	95.4	80.3	90.0
HFLAT(ours)	2025	90.8	95.9	<u>82.2</u>	<u>90.6</u>

with its momentum parameter set to 0.9 and the weight decay factor set to 1×10^{-4} . The initial learning rate is set to 8×10^{-3} and adjusted using cosine decay. The hyperparameter β controlling the balance in the loss function is set to 0.3. Finally, during the model training and evaluation phases, we set three distinct random seeds for each experiment and use the average of the results of three independent trials as the final result of each experiment. The values of key parameters, hardware and software requirements, and parameter count of our model are presented in Table 2.

4.3. Comparison with state-of-the-art methods

To evaluate the efficacy of HFLAT, we conduct comparative experiments against several state-of-the-art (SOTA) methods for occluded person ReID on the occlusion datasets Occluded-DukeMTMC and Occluded-ReID, as well as on the standard datasets Market-1501 and DukeMTMC-ReID.

Results on Occluded-DukeMTMC and Occluded-ReID: Table 3 summarizes the experimental results of our model compared to several SOTA occluded person ReID methods on the Occluded-DukeMTMC and Occluded-ReID datasets. Among these methods, PGFA (Miao et al., 2019), PVPM (Gao et al., 2020), HPNet (Huang et al., 2020), and POS (Wang et al., 2022a) are occluded person ReID methods based on external auxiliary information, while TransReID (He et al., 2021), PAT (Li et al., 2021), PFD (Wang et al., 2022b), FED (Wang et al., 2022c), SAP (Jia et al., 2023), PRE-Net (Yan et al., 2023), CAAO (Zhao et al., 2023), RFMT (Yang et al., 2023), SSPEM (Pang et al., 2024), MLRAT (Lin et al., 2024), OA-ReID (Ren et al., 2025) and DPEFormer (Zhang et al., 2025) are Transformer-based occluded person ReID methods. As depicted in Table 3, HFLAT achieves the mAP and Rank-1 accuracy of 64.7% and 79.6% on the Occluded-DukeMTMC dataset, as well as the mAP and Rank-1 accuracy of 84.9% and 89.8% on the Occluded-ReID dataset. Compared with the external auxiliary information-based methods, HFLAT improves the mAP and Rank-1 accuracy on the Occluded-DukeMTMC dataset by at least 10.7% and 14.6%, respectively, and the

mAP and Rank-1 accuracy on the Occluded-ReID dataset by at least 7.5% and 2.5%, respectively. Compared with the Transformer-based methods, HFLAT improves the mAP and Rank-1 accuracy by at least 1.6% and 5.4% on the Occluded-DukeMTMC dataset and by at least 1.3% and 0.9% on the Occluded-ReID dataset, respectively. It should be noted that, on the Occluded-ReID dataset, HFLAT's Rank-1 accuracy (89.8%) is only 0.9% higher than the second-best method MLRAT (Lin et al., 2024) (88.9%). HFLAT shows limited performance improvement on the Occluded-ReID dataset, likely due to the following two factors. First, every dataset has the performance ceiling effect. Existing advanced methods have achieved a high Rank-1 accuracy (over 87%) on the Occluded-ReID dataset. Given the fixed upper limit of Rank-1 accuracy (100%), the margin for improvement in Rank-1 accuracy will only become increasingly smaller. Second, the three core modules of our model-FHFM, FSM, and LFEM-are designed to enhance identity information in critical regions, suppress occlusion interference, and extract fine-grained local information, respectively. This enables our model to demonstrate certain advantages in scenarios with complex occlusions and significant viewpoint variations. Due to the relatively regular occlusion patterns in the Occluded-ReID dataset, our model fails to demonstrate significant advantages on this dataset.

Results on Market-1501 and DukeMTMC-ReID: Table 4 summarizes the experimental results of our model compared to several SOTA person ReID methods on the Market-1501 and DukeMTMC-ReID datasets. Among these methods, DSR (He et al., 2018a), PCB (Sun et al., 2018), PGFA (Miao et al., 2019), HOREID (Wang et al., 2020), HPNet (Huang et al., 2020), and POS (Wang et al., 2022a) are CNN-based methods, while TransReID (He et al., 2021), PAT (Li et al., 2021), FED (Wang et al., 2022c), FRT (Xu et al., 2022), FCFormer (Wang et al., 2024), RFMT (Yang et al., 2023), CTU (Zheng et al., 2024), OA-ReID (Ren et al., 2025) and DPEFormer (Zhang et al., 2025) are Transformer-based methods. As shown in Table 4, on the Market-1501 dataset, HFLAT achieves the mAP and Rank-1 accuracy of 90.8% and 95.9%, respectively, ranking first in performance. On the DukeMTMC-ReID dataset, HFLAT achieves the mAP and Rank-1 accuracy of 82.2% and 90.6%, although the mAP of HFLAT is slightly lower than that of RFMT (Yang et al., 2023), its Rank-1 accuracy exceeds that of RFMT (Yang et al., 2023). Additionally, while the Rank-1 accuracy of HFLAT is slightly lower than that of TransReID (He et al., 2021), its mAP surpasses that of TransReID (He et al., 2021). Moreover, Table 3 also shows that on both the Market-1501 and DukeMTMC-ReID datasets, HFLAT significantly outperforms CNN-based comparison methods in both mAP and Rank-1 accuracy. The experimental results additionally confirm the efficacy and advancement of HFLAT in occluded person ReID.

4.4. Ablation studies and hyperparameter analysis

4.4.1. Ablation studies

To assess the contribution of each module in HFLAT to its overall performance, we perform ablation studies on the Occluded-DukeMTMC, Occluded-ReID, Market-1501, and DukeMTMC-ReID datasets. We use TransReID (He et al., 2021) without the patch embedding module as the baseline, and sequentially add our designed modules to the baseline to assess their contribution to the model's performance. Specifically, we construct five models: Baseline + FHFM, Baseline + FSM, Baseline + LFEM, Baseline + FHFM + FSM, and Baseline + FHFM + FSM + LFEM, where, FHFM, FSM, and LFEM represent the feature hierarchy fusion module, the feature separation module, and the local feature extraction module, respectively. Table 5 presents the outcomes of the ablation experiments. In particular, the results from the original TransReID (He et al., 2021) are presented in Tables 3 and 4.

Effectiveness of the Feature Hierarchy Fusion Module (FHFM): As shown in Table 5, on the Occluded-DukeMTMC, Occluded-ReID, Market-1501 and DukeMTMC-ReID datasets, compared to the Baseline, the mAP of Baseline + FHFM improves from 48.8%, 77.8%, 87.7%, and 75.9% to 58.6%, 80.6%, 88.6%, and 79.6%, respectively, and its Rank-

1 accuracy increases from 66.0%, 81.5%, 94.8%, and 86.0% to 73.6%, 85.8%, 95.1%, and 88.4%, respectively. This indicates that FHFM enhances the discriminability of pedestrian features by hierarchically fusing image patch feature vectors.

Effectiveness of the Feature Separation Module (FSM): As shown in Table 5, on the Occluded-DukeMTMC, Occluded-ReID and DukeMTMC-ReID dataset, compared to the Baseline, the mAP of Baseline + FSM improves from 48.8%, 77.8%, and 75.9% to 52.1%, 78.7%, and 76.6%, respectively, and its Rank-1 accuracy increases from 66.0%, 81.5%, and 86.0% to 69.3%, 83.0%, and 86.6%, respectively. On the Market-1501 dataset, compared to the Baseline, the mAP of Baseline + FSM improves from 87.7% to 88.0%, and its Rank-1 accuracy remains unchanged. This demonstrates that FSM effectively reduces the interference of backgrounds and occlusions by separating foreground features and background features.

Effectiveness of the Local Feature Extraction Module (LFEM): As shown in Table 5, on the Occluded-DukeMTMC, Occluded-ReID, Market-1501 and DukeMTMC-ReID datasets, compared to the Baseline, Baseline + LFEM achieves improvements in the mAP from 48.8%, 77.8%, 87.7%, and 75.9% to 55.1%, 79.3%, 88.4%, and 78.3%, respectively, and its Rank-1 accuracy increases from 66.0%, 81.5%, 94.8%, and 86.0% to 73.0%, 83.5%, 95.0%, and 87.7%, respectively. This suggests that LFEM effectively enhances the ability of the model to capture fine-grained information, thereby improving its re-identification performance.

Joint Effectiveness of Modules: As illustrated in Table 5, on the Occluded-DukeMTMC dataset, Baseline + FHFM + FSM achieves the mAP and Rank-1 accuracy of 61.6% and 76.7%, respectively, significantly outperforming Baseline + FHFM and Baseline + FSM; Baseline + FHFM + LFEM achieves the mAP and Rank-1 accuracy of 62.5% and 76.8%, respectively, markedly outperforming Baseline + FHFM and Baseline + LFEM; Baseline + FSM + LFEM achieves the mAP and Rank-1 accuracy of 59.2% and 74.5%, respectively, markedly outperforming Baseline + FSM and Baseline + LFEM. On the Occluded-ReID dataset, Baseline + FHFM + FSM achieves the mAP and Rank-1 accuracy of 85.5% and 88.0%, respectively, markedly outperforming Baseline + FHFM and Baseline + FSM; Baseline + FHFM + LFEM achieves the mAP and Rank-1 accuracy of 83.9% and 88.5%, respectively, significantly outperforming Baseline + FHFM and Baseline + LFEM; Baseline + FSM + LFEM achieves the mAP and Rank-1 accuracy of 82.4% and 86.7%, respectively, markedly outperforming Baseline + FSM and Baseline + LFEM. On the Market-1501 dataset, Baseline + FHFM + FSM achieves the mAP and Rank-1 accuracy of 89.3% and 95.2%, respectively, markedly outperforming Baseline + FHFM and Baseline + FSM; Baseline + FHFM + LFEM achieves the mAP and Rank-1 accuracy of 90.3% and 95.5%, respectively, significantly outperforming Baseline + FHFM and Baseline + LFEM; Baseline + FSM + LFEM achieves the mAP and Rank-1 accuracy of 89.9% and 95.3%, respectively, markedly outperforming Baseline + FSM and Baseline + LFEM. On the DukeMTMC-ReID dataset, Baseline + FHFM + FSM achieves the mAP and Rank-1 accuracy of 81.3% and 90.5%, respectively, markedly outperforming Baseline + FHFM and Baseline + FSM; Baseline + FHFM + LFEM achieves the mAP and Rank-1 accuracy of 81.6% and 90.1%, respectively, significantly outperforming Baseline + FHFM and Baseline + LFEM; Baseline + FSM + LFEM achieves the mAP and Rank-1 accuracy of 80.5% and 90.1%, respectively, markedly outperforming Baseline + FSM and Baseline + LFEM. These experimental results demonstrate that every pairwise combination of the three modules effectively enhances the performance of our model. Furthermore, as illustrated in Table 5, on the Occluded-DukeMTMC dataset, Baseline + FHFM + FSM + LFEM achieves the best performance with the mAP and Rank-1 accuracy of 64.7% and 79.6%, respectively; On the Occluded-ReID dataset, Baseline + FHFM + FSM + LFEM achieves the best performance with the mAP and Rank-1 accuracy of 84.9% and 89.8%, respectively; On the Market-1501 dataset, Baseline + FHFM + FSM + LFEM also achieves the best performance with the mAP and Rank-1 accuracy of 90.8% and 95.9%, respectively; On the

Table 5

Results of ablation experiments (including mAP (%) and Rank-1 accuracy (%)) on the Occluded-DukeMTMC, Occluded-ReID, Market-1501 and DukeMTMC-ReID datasets.

Model	Occluded-DukeMTMC		Occluded-ReID		Market1501		DukeMTMC-ReID	
	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
Baseline	48.8	66.0	87.7	94.8	87.7	94.8	87.7	94.8
Baseline + FHFM	58.6	73.6	88.6	95.1	88.6	95.1	88.6	95.1
Baseline + FSM	52.1	69.3	88.0	94.8	88.0	94.8	88.0	94.8
Baseline + LFEM	55.1	73.0	88.4	95.0	88.4	95.0	88.4	95.0
Baseline + FHFM + FSM	61.6	76.7	89.3	95.2	89.3	95.2	89.3	95.2
Baseline + FHFM + LFEM	62.5	76.8	90.3	95.5	90.3	95.5	90.3	95.5
Baseline + FSM + LFEM	59.2	74.5	89.9	95.3	89.9	95.3	89.9	95.3
Baseline + FHFM + FSM + LFEM	64.7	79.6	90.8	95.9	90.8	95.9	90.8	95.9

Table 6

Experimental results(including mAP (%) and Rank-1 accuracy (%)) for the number of levels in patch sequence segmentation in FHFM on the Occluded-DukeMTMC, and Occluded-ReID datasets.

Model	Occluded-DukeMTMC		Occluded-ReID	
	mAP	Rank-1	mAP	Rank-1
no segmentation	59.2	74.5	82.4	86.7
Segmenting the patch sequence into two different levels of subsequences	61.1	77.3	83.2	88.1
Segmenting the patch sequence into three different levels of subsequences	63.2	78.6	84.0	89.3
Segmenting the patch sequence into four different levels of subsequences	64.7	79.6	84.9	89.8
Segmenting the patch sequence into five different levels of subsequences	63.4	79.5	83.9	88.9

Table 7

Experimental results(including mAP (%) and Rank-1 accuracy (%)) for the number of levels in patch sequence segmentation in FHFM on the Market-1501, and DukeMTMC-ReID datasets.

Model	Market-1501		DukeMTMC-ReID	
	mAP	Rank-1	mAP	Rank-1
no segmentation	89.9	95.3	80.5	89.1
Segmenting the patch sequence into two different levels of subsequences	90.3	95.5	80.9	89.5
Segmenting the patch sequence into three different levels of subsequences	90.5	95.6	81.6	90.1
Segmenting the patch sequence into four different levels of subsequences	90.8	95.9	82.2	90.6
Segmenting the patch sequence into five different levels of subsequences	90.0	95.3	81.0	89.7

Occluded-DukeMTMC dataset, Baseline + FHFM + FSM + LFEM achieves the best performance with the mAP and Rank-1 accuracy of 82.2% and 90.6%, respectively. This indicates that the synergistic effect of the three modules significantly improves the performance of our model.

4.4.2. Hyperparameter analysis

To validate the impact of the number of levels in patch sequence segmentation in FHFM on the performance of our model, we conduct ablation experiments on the Occluded-DukeMTMC, Occluded-ReID, Market-1501, and DukeMTMC-ReID datasets. Specifically, we establish five distinct patch sequence segmentation approaches, i.e., no segmentation, and segmenting the patch sequence into two, three, four, or five different levels of subsequences. The experimental results are presented in Tables 6 and 7.

As shown in Tables 6 and 7, our model achieves optimal performance when the patch sequence is segmented into four different levels of subsequences. Specifically, on the Occluded-DukeMTMC dataset, compared to no segmentation and segmenting the patch sequence into two, three, and five different levels of subsequences, segmenting the patch sequence into four different levels of sub-sequences, the mAP of our model is improved by 5.5%, 3.6%, 1.5%, and 1.3%, respectively, and its Rank-1 accuracy is improved by 5.1%, 2.3%, 1.0%, and 0.1%, respectively. On the Occluded-ReID dataset, compared to no segmentation and segmenting the patch sequence into two, three, and five different levels of subsequences, segmenting the patch sequence into four different levels of sub-sequences, the mAP of our model is improved by 2.5%, 1.7%, 0.9%, and 1.0%, respectively, and its Rank-1 accuracy is improved by 3.1%, 1.7%, 0.5%, and 0.9%, respectively. On the Market-1501 dataset, compared to no segmentation and segmenting the patch sequence into two, three, and five different levels of subsequences, segmenting the patch sequence into four different levels of sub-sequences, the mAP of our

model is improved by 0.9%, 0.5%, 0.3%, and 0.8%, respectively, and its Rank-1 accuracy is improved by 0.6%, 0.4%, 0.3%, and 0.6%, respectively. On the DukeMTMC-ReID dataset, compared to no segmentation and segmenting the patch sequence into two, three, and five different levels of subsequences, segmenting the patch sequence into four different levels of sub-sequences, the mAP of our model is improved by 1.7%, 1.3%, 0.6%, and 1.2%, respectively, and its Rank-1 accuracy is improved by 1.5%, 1.1%, 0.5%, and 0.9%, respectively. Tables 6 and 7 also shows that, on each dataset, as the number of levels in patch sequence segmentation increases, our model's performance exhibits an initial improvement followed by a decline. This may be attributable to the fact that, with too few subsequences, our model is unable to adequately capture the differences in identity information across some key regions; conversely, with too many subsequences, our model fails to completely capture the identity information contained in some key regions. Moreover, from the perspective of the semantic structure of pedestrian images, the optimal number of levels is highly compatible with the characteristics of human anatomy. Pedestrian images inherently comprise four key regions with clearly defined semantic boundaries: the head, the upper body, the lower body, and the feet. These regions respectively contain information such as hairstyle and facial features, the style and color of upper garments, the style and color of trousers, and shoes. These elements constitute the key semantic information for person re-identification. If the number of levels is less than 4, FHFM will inevitably force the merging of adjacent semantic regions (such as the upper body with the lower body, or the lower body with the feet), thereby failing to accurately capture the distinctive features of different key regions. If the number of levels exceeds 4, FHFM will inevitably split a complete semantic region (such as dividing the upper body into the chest and the waist), thereby failing to obtain complete key semantic information.

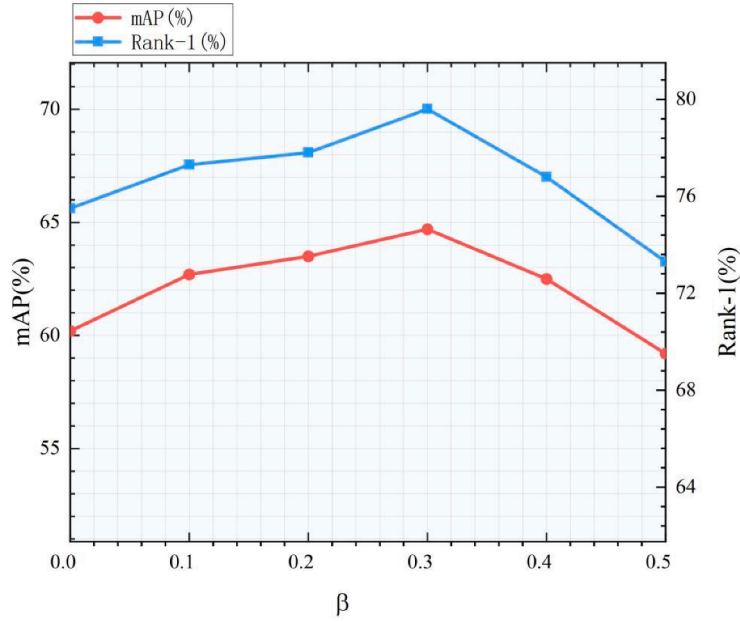


Fig. 6. Curve of the performance of HFLAT in relation to the dimension reduction ratio β .

Table 8

Significance testing results (including mAP (%), Rank-1 accuracy (%), test statistic t , and probability p) for HFLAT on the Occluded-DukeMTMC, Occluded-ReID, Market-1501, and DukeMTMC-ReID datasets.

seed	method	Occluded-DukeMTMC		Occluded-ReID		Market-1501		DukeMTMC-ReID	
		mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
1	Baseline	48.6	65.9	77.5	81.2	87.3	94.9	75.7	86.4
	ours	64.9	66.3	84.7	89.6	91.0	96.1	82.1	90.6
2	Baseline	48.9	66.3	77.9	81.7	87.9	95.0	76.2	85.8
	ours	64.5	79.4	85.1	89.9	90.9	96.0	82.2	90.8
3	Baseline	48.8	65.8	78.1	81.6	87.8	94.5	75.8	85.9
	ours	64.7	79.6	85.2	90.0	90.6	95.7	82.3	90.5
t		78.58	54.04	215.00	125.00	11.61	17.00	41.24	19.92
p		0.00008	0.00017	0.00001	0.00003	0.00367	0.00172	0.00029	0.00126

In our method, β serves as a balance coefficient in the loss function, and its value directly influences the extent to which our model focuses on the feature separation. To achieve accurate pedestrian re-identification, our model not only employs the ID loss and the triplet loss to ensure the discriminative power of the final pedestrian feature vectors, but also utilizes the foreground-background contrastive loss to ensure that the final pedestrian feature vectors contain as much foreground information as possible. To investigate the effects of the balance parameter β in the loss function on the overall performance of our model, we conduct experiments on the Occluded-DukeMTMC dataset, as illustrated in Fig. 6.

As seen in Fig. 6, as β increases, our model's mAP and Rank-1 accuracy all show an increasing and then decreasing trend, reaching the optimal performance when $\beta = 0.3$. This may be due to the following reasons. When β is set too high, the foreground-background contrastive loss is overly emphasized. This causes our model to excessively focus on the feature separation, thereby suppressing FHF's ability to extract identity information from key regions and LFEM's ability to capture fine-grained local details. When β is set too low, the foreground-background contrastive loss carries insufficient weight, preventing FSM from effectively separating foreground features and background features. Consequently, the foreground features output by FSM contain excessive background interference, thereby reducing the discriminative power of the final pedestrian feature vectors.

4.5. Statistical significance testing

To exclude the possibility that the performance improvement of HFLAT is attributable to random factors, we conduct experiments on the Occluded-DukeMTMC, Occluded-ReID, Market-1501, and DukeMTMC-ReID datasets for both the Baseline and our method with three random seeds, and perform paired t-tests to verify the statistical significance of the experimental results. Specifically, we firstly denote the three experimental results of our method on any dataset regarding a certain metric (mAP or Rank-1 accuracy) as X_1, X_2, X_3 , and denote the corresponding results of the Baseline as Y_1, Y_2, Y_3 . Then, we make $D_1 = X_1 - Y_1$, $D_2 = X_2 - Y_2$, and $D_3 = X_3 - Y_3$. Hence, D_1, D_2 , and D_3 are mutually independent and follow a same normal distribution $\mathcal{N}(\mu_D, \sigma_D^2)$. We set the null hypothesis H_0 as: $\mu_D \leq 0$, meaning the performance of our method is significantly inferior to that of the Baseline. We set the alternative hypothesis H_1 as: $\mu_D > 0$, meaning the performance of our method is significantly superior to that of the Baseline. The significance level α is set to 0.01. In Table 8, we present the test statistic t for each set of paired data and the corresponding probability p of accepting the null hypothesis H_0 . As shown in Table 8, for the Occluded-DukeMTMC, Occluded-ReID, Market-1501, and DukeMTMC-ReID datasets, the value of p for each set of paired data is below the significance level α . This indicates that we should reject the null hypothesis H_0 and accept the alternative hypothesis H_1 . In other words, the performance of our method is significantly superior to that of the Baseline.

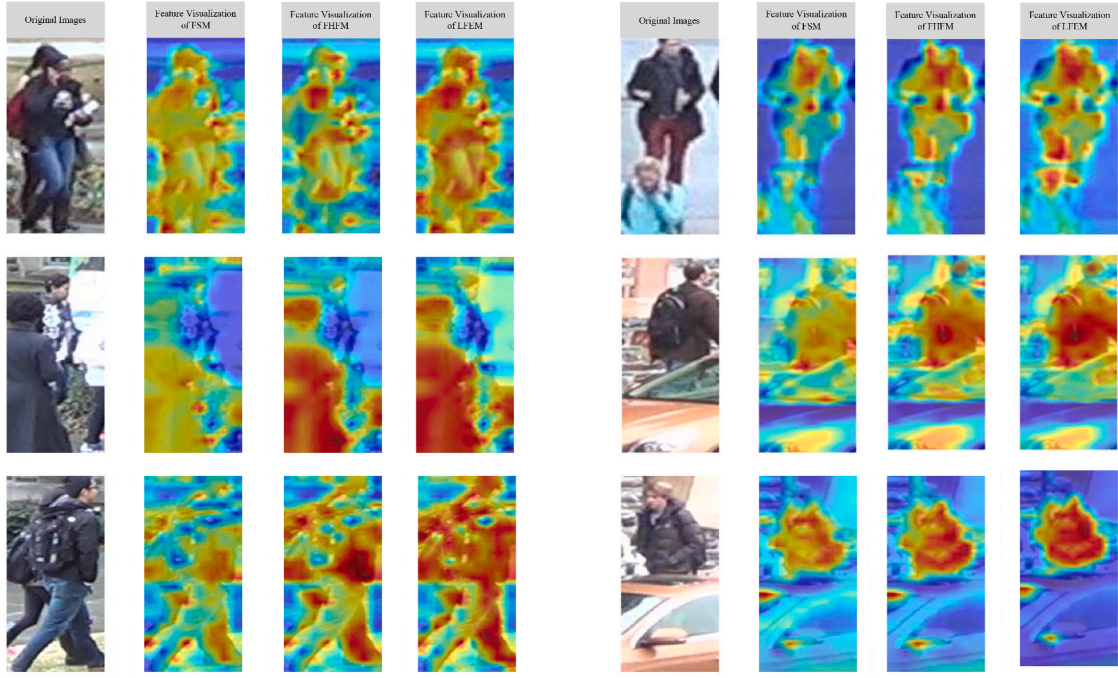


Fig. 7. Feature visualization of the three modules.

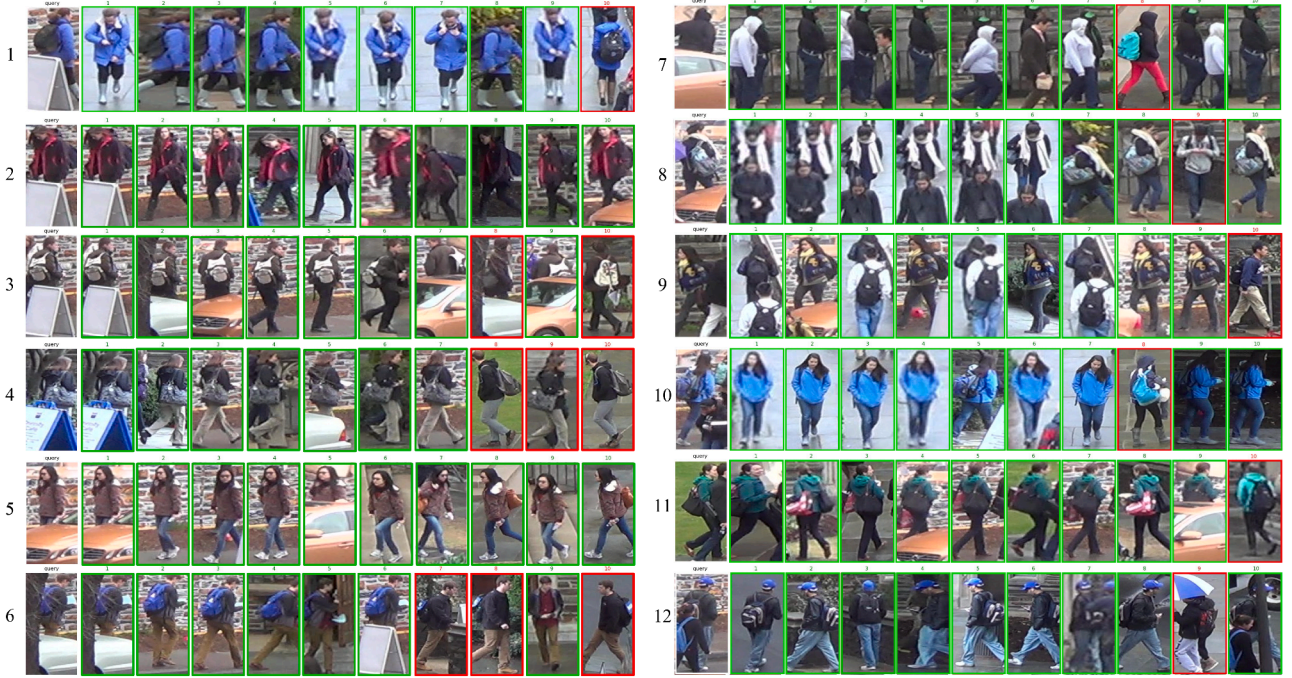


Fig. 8. Visualization of retrieval results.

4.6. Visualization of results

To demonstrate the effectiveness of HFLAT clearly, we visualize the features of FHFMM, FSM, and LFEM in Fig. 7. As shown in Fig. 7, FSM can separate foreground features and background features effectively, FHFMM can focus well on key regions, and LFEM can capture fine-grained local features effectively. Nevertheless, it is also evident from Fig. 7 that, when the occluding object is another pedestrian, all the three modules exhibit a tendency to focus on the interfering pedestrian. This may prevent our model from extracting effective discriminative features from the small, unobstructed body regions of the target pedestrian. In future

work, we will design a region-guided feature extraction strategy to enhance the performance of our model.

To visualize the effectiveness of HFLAT more clearly, we present the retrieval results for several query images from the Occluded-DukeMTMC dataset in Fig. 8. In Fig. 8, the query images in rows 1–4, 5–8, and 9–12 are occluded by billboards, vehicles, and pedestrians, respectively. The image on the leftmost of each row is the query image, while the remaining ten images are the retrieved results sorted in descending order of similarity to the query image. Green numbers and green bounding boxes indicate the accurate retrieval results, while red numbers and red bounding boxes denote the inaccurate retrieval results. As shown in Fig. 8,

HFLAT effectively retrieves the top 6 images belonging to the same individual as the corresponding query image. Nevertheless, as shown in Fig. 8, for some query images, incorrect retrieval results still exist, although they all appear latter positions. This may be because, in these images, the target pedestrians are severely occluded, and our model fails to effectively extract discriminative features from the excessively small un-occluded body regions. In future work, we may design a local-guided feature extraction strategy to boost our model's performance.

5. Conclusions

This paper proposes a hierarchical fusion and local-aware Transformer (HFLAT) for occluded person ReID. In HFLAT, we first use a feature hierarchical fusion module to hierarchize and fuse the image patch feature vector sequence, enhancing the identity discriminative features of key regions. Then, we use a feature separation module to separate the foreground features and background features, reducing the negative impact of backgrounds and occlusions on the model's performance. Additionally, we use a local feature extraction module to capture the fine-grained local features from different regions of pedestrian images. Extensive experimental results on four datasets not only validate the effectiveness of our model but also demonstrate its generalization capability, robustness, and adaptability. However, due to limitations in our research conditions, we did not analyze the impact of different occlusion types on our model's performance, nor did we evaluate its performance in real-world complex environments or under more severe occlusion conditions. In our future work, if the conditions for our research improve, we will attempt to conduct an extended analysis of how different occlusion types affect the performance of our model. We will also attempt to evaluate the generalization capability, robustness and adaptability of our model in real-world complex environments and under more severe occlusion conditions. Furthermore, we will try to employ strategies such as multimodal information fusion, optimization of key region localization, extraction of discriminative information from non-key regions, and enhancement of feature diversity to improve our model's performance in extreme occlusion scenarios. In particular, although the ablation experimental results on the four datasets demonstrate the effectiveness of LFEM, its equal height-based image division strategy inevitably fails to properly align body parts in extreme poses (e.g., sitting or crouching). We will design an adaptive pedestrian image segmentation strategy based on pose-aware attention mechanisms to address this issue in our future work. Moreover, we will try to optimize our model to reduce its computational complexity.

CRedit authorship contribution statement

Haishun Du: Writing – review & editing, Writing – original draft, Supervision, Funding acquisition; **Chuaner Huang:** Writing – review & editing, Software, Data curation; **Linbing Cao:** Writing – original draft, Methodology; **Jieru Li:** Validation, Formal analysis; **Wenzhe Zhang:** Visualization, Validation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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